

Swim Bladder: 8-Thruster Central Star-Node PDB

Technical Specification for 8× Blue Robotics T200 + Basic ESC System

Power Source: 4S battery direct input, 14.8 V nominal / 16.8 V fully charged

Architecture: Cable-side main fuse, dual XT90 input, one compact central high-current node, eight fused ESC branches

Contents

1	Purpose of This Architecture	2
2	Electrical Assumptions	2
3	High-Level System Architecture	3
3.1	Recommended Power Tree	3
3.2	Core Layout Principle	3
4	Recommended Physical Layout	3
4.1	Top-Level Board Layout	3
4.2	Central Node Detail	3
4.3	Ground Layout	4
5	Fuse Architecture	4
5.1	Fuse Philosophy	4
5.2	Cable-Side Main Fuse	4
5.3	Branch Fuse Sizing	4
6	PCB Stack-Up	5
7	Copper Geometry and Current Carrying Design	5
7.1	Estimation Method	5
7.2	Branch Width	5
7.3	Central Node Size	6
7.4	Input Connector Sharing	6
8	Voltage Drop and Loss Estimates	6
9	Via Specification	7
9.1	Recommended Via Dimensions	7
9.2	Recommended Via Counts	7
10	Connector Recommendations	7
10.1	Main Battery Input	7
10.2	ESC Output Connectors	7
11	Passive Components and Protection	8
11.1	Capacitors	8
11.2	TVS Protection	8
11.3	Bleeder Resistor	9
11.4	Power Indicator LED	9
12	PWM and Signal Architecture	9

13 Recommended BOM	9
13.1 Power Path BOM	9
13.2 Passive and Protection BOM	9
13.3 Signal BOM	9
14 Layout Checklist	9
15 Test Plan	10
15.1 Continuity Test	10
15.2 Low-Current Power-Up	11
15.3 Progressive Load Test	11
16 Final Design Summary	11

1 PURPOSE OF THIS ARCHITECTURE

This document describes a compact high-current power distribution board for an 8-thruster ROV system using eight Blue Robotics T200 thrusters and eight Basic ESCs powered directly from a 4S lithium battery. The previous dual-bank architecture was useful for larger boards, but for a compact board of approximately 90 mm by 90 mm, the two bank nodes physically merge into the same central copper region.

The recommended compact architecture is therefore a **single central star node**: the battery input feeds a very short, wide, heavily via-stitched copper island, and all eight branch fuses are placed close to that node.

Recommended compact architecture:

```
Battery -> cable-side main fuse -> e-stop/contactator -> dual XT90 input
        -> short central 200 A node
        -> eight 30 A branch fuses
        -> eight XT60 ESC outputs
```

Design Note: The board is no longer treated as two separated 4-thruster banks. Instead, it is treated as one compact star PDB. This is acceptable because the main fuse is moved upstream on the cable and the high-current PCB region is extremely short.

Warning: This architecture depends on having a proper upstream main fuse near the battery. Without a main cable-side fuse, the central node and PCB input copper before the branch fuses would be insufficiently protected.

2 ELECTRICAL ASSUMPTIONS

A 4S lithium battery is assumed:

$$V_{\text{nominal}} \approx 14.8 \text{ V} \quad (1)$$

$$V_{\text{full}} = 4 \times 4.2 \text{ V} = 16.8 \text{ V} \quad (2)$$

Use conservative T200 current at the 16 V class:

$$I_{\text{thruster,max}} \approx 25 \text{ A} \quad (3)$$

For eight thrusters:

$$I_{\text{total,max}} = 8 \times 25 \text{ A} = 200 \text{ A} \quad (4)$$

Recommended system design target:

$$I_{\text{central node,continuous design}} = 200 \text{ A} - 220 \text{ A} \quad (5)$$

$$I_{\text{central node,burst design}} = 280 \text{ A} - 300 \text{ A} \quad (6)$$

$$I_{\text{branch,design}} = 30 \text{ A} \quad (7)$$

Per-thruster electrical power at 16 V:

$$P_{\text{thruster}} = VI = 16 \text{ V} \times 25 \text{ A} = 400 \text{ W} \quad (8)$$

Whole system:

$$P_{\text{system}} = 8 \times 400 \text{ W} = 3200 \text{ W} \quad (9)$$

3 HIGH-LEVEL SYSTEM ARCHITECTURE

3.1 Recommended Power Tree

```

4S Battery, 14.8 V nominal / 16.8 V full
|
[Main fuse on cable, near battery: 225-250 A typical]
|
[Electrical E-stop / DC contactor, optional but recommended]
|
[Dual XT90-S or dual XT90 input to PCB]
|
[Short central 200 A star node]
|
|-- 30 A fuse -- XT60 -- ESC1 -- T200
|-- 30 A fuse -- XT60 -- ESC2 -- T200
|-- 30 A fuse -- XT60 -- ESC3 -- T200
|-- 30 A fuse -- XT60 -- ESC4 -- T200
|-- 30 A fuse -- XT60 -- ESC5 -- T200
|-- 30 A fuse -- XT60 -- ESC6 -- T200
|-- 30 A fuse -- XT60 -- ESC7 -- T200
|-- 30 A fuse -- XT60 -- ESC8 -- T200

```

3.2 Core Layout Principle

The central node architecture minimizes PCB area by reducing the length of copper carrying full system current.

```

Bad compactness:
Battery input -> long 200 A spine -> ESC1 -> ESC2 -> ... -> ESC8

Good compactness:
Battery input -> short 200 A copper node -> eight short 30 A branches

```

Warning: The central node should be designed as a copper bus plate built into the PCB, not as a normal routed trace. It must be short, wide, multi-layered, heavily via-stitched, and thermally tested.

4 RECOMMENDED PHYSICAL LAYOUT

4.1 Top-Level Board Layout

For a compact 90 mm by 90 mm PCB, place the dual XT90 input close to the center or edge-center of the board, then place the eight branch fuse inputs as close as possible around the central node.

Top view concept, compact central-node version:

```

      DUAL XT90 BATTERY INPUT
      XT90-A XT90-B
        \ /
         \ /
          \ /
+-----+
| CENTRAL 200 A NODE |
+-----+
 / / / | \ \ \
F1 F2 F3 F4 F5 F6 F7/F8
| | | | | |
X1 X2 X3 X4 X5 X6 X7/X8
| | | | | |
E1 E2 E3 E4 E5 E6 E7/E8

```

X1-X8 = XT60 branch outputs
E1-E8 = ESC power leads
PWM IDC connector should sit near a quiet board edge.

4.2 Central Node Detail

The central node is a compact copper island connecting the dual XT90 positive inputs to all eight branch fuse inputs.

```

Dual XT90 positive pads
  | |
+-----+
| 200 A NODE | short, wide, via-stitched copper island
+-----+
 / / / | \ \ \
F1 F2 F3 F4 F5 F6 F7 F8
| | | | | |
XT60 branch outputs to ESCs

```

Design Note: The branch split should happen immediately after the dual XT90 input. The central node should feed all fuse input pads directly with no thin neck-downs.

4.3 Ground Layout

The positive side uses a central star node and individual branch fuses. The ground side should be a common, low-impedance return system.

```
Battery - from dual XT90 input
|
+-- common GND plane / GND node
|   |-- XT60 negative return for ESC1-ESC8
|   '-- flight controller signal ground reference
```

Design Note: Use common GND across the board. ESC PWM signals require a common reference to the flight controller. Do not isolate grounds unless there is a specific measurement reason.

5 FUSE ARCHITECTURE

5.1 Fuse Philosophy

Recommended protection hierarchy:

- main fuse on cable, close to the battery,
- optional contactor or electrical e-stop before the PCB,
- one branch fuse per ESC on the PCB.

5.2 Cable-Side Main Fuse

Because the PCB no longer contains bank fuses, the upstream cable-side main fuse becomes mandatory.

Recommended:

$$I_{\text{main fuse}} = 225 \text{ A} - 250 \text{ A} \quad (10)$$

Placement:

$$\text{Battery} + \rightarrow \text{main fuse} \rightarrow \text{e-stop/contactor} \rightarrow \text{PCB input} \quad (11)$$

Recommended fuse type:

- ANL fuse,
- MEGA fuse,
- high-current marine or EV-style battery fuse holder.

Warning: The main fuse must be close to the battery, not only close to the PCB. Its job is to protect the battery cable, e-stop wiring, PCB input, and central node from catastrophic shorts before the branch fuses.

5.3 Branch Fuse Sizing

Each ESC branch:

$$I_{\text{branch actual}} \approx 24 \text{ A} - 25 \text{ A} \quad (12)$$

$$I_{\text{ESC class}} \approx 30 \text{ A} \quad (13)$$

Recommended:

$$I_{\text{branch fuse}} = 30 \text{ A} \quad (14)$$

Recommended fuse holder:

- Keystone 3568 mini blade fuse holder, PCB-mounted, or
- Littelfuse FLR 178.6165 ATO holder with caution, or
- off-board inline ATO/ATC fuse holder for better thermal margin.

Warning: Do not increase branch fuses above 30 A unless the ESC, wiring, connector, and thermal path are explicitly validated. If nuisance blowing occurs, investigate throttle limiting, cooling, and fuse time-current curves first.

6 PCB STACK-UP

Recommended 4-layer stack-up:

Table 1: Recommended 4-Layer Stack-Up

Layer	Function
L1 / F.Cu	+VBAT central node and branch fuse input/output pours
L2 / In1.Cu	GND high-current return plane
L3 / In2.Cu	+VBAT central node and branch pours, stitched to L1
L4 / B.Cu	GND plane + PWM / telemetry / low-current signal routing

$$+VBAT \text{ path : } L1 + L3 \quad (15)$$

$$GND \text{ path : } L2 + L4 \quad (16)$$

Design Note: Use L1 and L3 in parallel for the central +VBAT node and branch paths. Use L2 and L4 in parallel for the common GND return.

7 COPPER GEOMETRY AND CURRENT CARRYING DESIGN

7.1 Estimation Method

Use IPC-style estimation:

$$I = k\Delta T^b A^c \quad (17)$$

Rearranged:

$$A = \left(\frac{I}{k\Delta T^b} \right)^{1/c} \quad (18)$$

Common constants:

$$k_{\text{external}} = 0.048, \quad k_{\text{internal}} = 0.024, \quad b = 0.44, \quad c = 0.725 \quad (19)$$

For 4 oz copper:

$$t_{\text{Cu}} \approx 140 \mu\text{m} \quad (20)$$

7.2 Branch Width

Each branch is designed for:

$$I_{\text{branch design}} = 30 \text{ A} \quad (21)$$

Estimated minimum 4 oz external copper width at approximately 20 °C rise:

$$W_{30A} \approx 5.4 \text{ mm} \quad (22)$$

Recommended practical branch pour width:

$$W_{\text{branch}} = 12 \text{ mm} - 15 \text{ mm} \quad (23)$$

Absolute compact minimum for very short branch routes:

$$W_{\text{branch, min}} \approx 8 \text{ mm} - 10 \text{ mm} \quad (24)$$

7.3 Central Node Size

The central node carries full system current only over a very short distance.

Central node design target:

$$I_{\text{node,max}} = 200 \text{ A} \quad (25)$$

$$I_{\text{node,design}} = 200 \text{ A} - 220 \text{ A} \quad (26)$$

$$I_{\text{node,burst}} = 280 \text{ A} - 300 \text{ A} \quad (27)$$

For a long one-layer 4 oz trace, 200 A would require a very large width. However, this board uses a compact multi-layer copper island rather than a long trace. The design goal is therefore:

Table 2: Recommended Central Node Geometry

Parameter	Recommendation
Node type	Large compact copper polygon, not a routed trace
Node location	Immediately after dual XT90 input pads
Node layers	L1 + L3 parallel + VBAT copper
Node length	As short as possible; preferably below 20 mm–30 mm before branch split
Node area	Maximize available area around XT90 and fuse inputs
Minimum effective copper section	Avoid any local neck-down below approximately 50 mm equivalent where full current flows
Via stitching	Dense via field across the node and around every input/fuse pad
Branch exits	Eight direct short branch pours from node to branch fuses
Copper exposure	Optional exposed copper/solder reinforcement on central node if fabrication allows

Warning: Do not draw a thin trace from the XT90 input to the fuse inputs. The central node should be a filled polygon surrounding the input pads and fuse input pads.

7.4 Input Connector Sharing

Using dual XT90 inputs helps share current between two physical connectors. The two XT90 positive pads should join immediately into the central node. The two XT90 negative pads should join immediately into the common GND plane.

$$I_{\text{per XT90,ideal}} \approx \frac{200 \text{ A}}{2} = 100 \text{ A} \quad (28)$$

Warning: Current sharing between two connectors is not automatically perfect. Keep both input paths symmetrical, short, and wide. Use the same wire gauge and length for both XT90 feeds if possible.

8 VOLTAGE DROP AND LOSS ESTIMATES

For a short central node, losses can be manageable despite high current. Example: 50 mm effective-width, 140 μm thick, 25 mm long copper section on one layer:

$$A = 0.050 \times 0.000140 = 7.0 \times 10^{-6} \text{ m}^2 \quad (29)$$

$$R = \frac{1.724 \times 10^{-8} \times 0.025}{7.0 \times 10^{-6}} \quad (30)$$

$$\approx 0.0000616 \text{ } \Omega = 0.0616 \text{ m}\Omega \quad (31)$$

At 200 A:

$$V_{\text{drop}} = IR = 200 \times 0.0000616 \approx 0.012 \text{ V} \quad (32)$$

$$P_{\text{loss}} = I^2 R = 200^2 \times 0.0000616 \approx 2.46 \text{ W} \quad (33)$$

With L1 and L3 sharing current, effective resistance can be lower if the vias are dense and current spreads well. In practice, pad bottlenecks, solder joints, fuse holders, and connector interfaces may dominate heating.

Design Note: The central-node layout reduces heating mainly by reducing the length of copper that carries full system current. After the node splits, each branch only carries approximately 25 A–30 A.

9 VIA SPECIFICATION

9.1 Recommended Via Dimensions

Table 3: Recommended Via Dimensions

Parameter	Recommended	Notes
Via drill	0.5 mm–0.6 mm	Larger preferred if fab allows
Via diameter	1.0 mm–1.2 mm	Use for high-current stitching
Same-net via pitch	1.5 mm–2.5 mm	Use arrays across node and branches
Different-net clearance	≥ 1.0 mm preferred	Especially near node and connector regions

9.2 Recommended Via Counts

Table 4: Via Counts for Compact Central-Node Design

Location	+VBAT Vias	GND Vias	Notes
Dual XT90 input region	80+ total	80+ total	Around both positive and negative input pads
Central 200 A node	100+ distributed	100+ nearby GND stitching	Dense via field across the node
Each branch exit from node	8–12	8–12	Per branch exit
Each ESC branch	12–20	12–20	Per branch
Each XT60 connector region	12–16	12–16	Prevent pad bottlenecks
Capacitor region	6–10	6–10	Short path to planes
TVS region	12–20	12–20	Low-impedance clamp path

Warning: Distribute vias across the node. Do not place only one via cluster; the goal is current spreading between L1 and L3 across the whole copper node.

10 CONNECTOR RECOMMENDATIONS

10.1 Main Battery Input

For compactness, the current design uses dual XT90 input connectors. This is acceptable if the upstream wiring is symmetric and the main fuse is on the cable.

Table 5: Main Input Connector Options

Connector	Verdict
Dual XT90-S	Good compact option if current shares symmetrically.
QS8-S	Preferred single high-current connector if packaging allows.
AS150 or AS150U	Good high-current anti-spark option.
M6 or M8 ring lug / stud input	Most mechanically robust option, but larger.
Single XT90-S	Not preferred for full 8-thruster 4S operation.

10.2 ESC Output Connectors

At 4S, each ESC branch is expected to operate around 25 A and should be designed around 30 A. XT60 is preferred for each branch output.

Table 6: ESC Output Connector Options

Connector	Verdict
XT60	Preferred for branch-level margin and lower connector heating.
XT30	Borderline; use only if short, well-soldered, cooled, and thermally validated.
Direct terminal, ring, or spade	Mechanically robust if packaging allows.

Design Note: Horizontal XT60 connectors may be placed directly beside the central node area, but each XT60 must be behind its own branch fuse. Do not feed multiple ESCs through one XT60.

11 PASSIVE COMPONENTS AND PROTECTION

11.1 Capacitors

Recommended capacitor placement follows the central-node architecture:

- main bulk capacitor near the dual XT90 input or central node,
- branch bulk capacitors near each XT60/ESC output,
- ceramic capacitors near branch outputs for high-frequency decoupling.

Main node capacitance:

$$C_{\text{main bulk}} = 2200\ \mu\text{F} - 4700\ \mu\text{F},\ 35\ \text{V or } 50\ \text{V} \quad (34)$$

Space-saving alternatives:

$$C_{\text{main alternative}} = 2 \times 2200\ \mu\text{F or } 3 \times 1500\ \mu\text{F} \quad (35)$$

Per ESC branch:

$$C_{\text{branch bulk}} = 470\ \mu\text{F} - 1000\ \mu\text{F},\ 35\ \text{V or } 50\ \text{V} \quad (36)$$

$$C_{\text{HF}} = 1\ \mu\text{F},\ 50\ \text{V},\ X7R \quad (37)$$

$$C_{\text{decouple}} = 100\ \text{nF},\ 50\ \text{V},\ X7R \quad (38)$$

Recommended compact capacitor footprints:

- 1000 μF , 35 V: `Capacitor_THT:CP_Radial_D8.0mm_P3.50mm` if exact part fits, otherwise `D10.0mm_P5.00mm`;
- 4700 μF , 35 V: `Capacitor_THT:CP_Radial_D12.5mm_P5.00mm` if exact part fits;
- ceramic capacitors: `Capacitor_SMD:C_1206_3216Metric` preferred.

11.2 TVS Protection

For a 4S pack, the TVS must not conduct during normal fully charged battery voltage of 16.8 V. Recommended:

$$V_{\text{TVS standoff}} = 22\ \text{V} - 24\ \text{V} \quad (39)$$

Recommended part class:

- SMCJ24A or similar for compact protection,
- SM8S24A or similar for stronger surge margin if space allows.

KiCad symbol and footprint:

- symbol: `Device:D_TVS`,
- footprint: `Diode_SMD:D_SMC` for SMCJ class.

Wiring:

$$\text{TVS cathode} \rightarrow +V_{\text{BAT}} \quad (40)$$

$$\text{TVS anode} \rightarrow GND \quad (41)$$

Warning: The TVS should be close to the input/central node, but it must not create a narrow bottleneck in the main current path. Prioritize a clean central copper node.

11.3 Bleeder Resistor

Recommended:

$$R_{\text{bleeder}} = 10\text{ k}\Omega\text{--}47\text{ k}\Omega \quad (42)$$

$$P_{\text{rating}} = 0.25\text{ W minimum} \quad (43)$$

At 16.8 V and 10 k Ω :

$$P = \frac{16.8^2}{10000} = 0.028\text{ W} \quad (44)$$

11.4 Power Indicator LED

LED resistor at full charge for approximately 2 mA:

$$R = \frac{16.8 - 2}{0.002} = 7.4\text{ k}\Omega \quad (45)$$

Recommended:

$$R_{\text{LED}} = 6.8\text{ k}\Omega\text{--}10\text{ k}\Omega, 0.25\text{ W} \quad (46)$$

12 PWM AND SIGNAL ARCHITECTURE

PWM is low-current and distributed through one 16-pin IDC connector.

Table 7: 16-Pin IDC PWM Pinout

Pin	Signal	Pin	Signal
1	PWM1	2	GND
3	PWM2	4	GND
5	PWM3	6	GND
7	PWM4	8	GND
9	PWM5	10	GND
11	PWM6	12	GND
13	PWM7	14	GND
15	PWM8	16	GND

Per PWM channel:

$$R_{\text{series}} = 100\text{ }\Omega\text{--}220\text{ }\Omega \quad (47)$$

$$R_{\text{pulldown}} = 10\text{ k}\Omega \quad (48)$$

Recommended:

$$R_{\text{series}} = 100\text{ }\Omega, \quad R_{\text{pulldown}} = 10\text{ k}\Omega \quad (49)$$

Design Note: Keep the IDC connector near a quiet board edge. Route PWM away from the dual XT90 input, TVS diode, fuse input pads, and central node.

13 RECOMMENDED BOM

13.1 Power Path BOM

13.2 Passive and Protection BOM

13.3 Signal BOM

14 LAYOUT CHECKLIST

Table 8: Power Path Components for Compact Central-Node Design

Ref.	Component	Rating / Value	Qty
J-IN-A/B	Main input connectors	Dual XT90-S or dual XT90	2
F-MAIN	Cable-side main fuse	225 A–250 A, DC-rated	1 off-PCB
K-ESTOP	Optional contactor / e-stop	200 A+ DC contactor, 12 V coil typical	1 off-PCB
J-ESC-1–8	ESC output connector	XT60 preferred; XT30 derated	8
F-ESC-1–8	Branch fuse	30 A mini blade / ATO / ATC	8

Table 9: Passive and Protection Components

Ref.	Component	Rating / Value	Qty
C-MAIN	Main bulk capacitor	2200 μ F–4700 μ F, 35 V or 50 V	1–2
C-ESC-1–8	Branch electrolytic capacitor	470 μ F–1000 μ F, 35 V or 50 V	8
C-HF-1–8	Ceramic capacitor	1 μ F, 50 V, X7R	8
C-DEC-1–8	Ceramic capacitor	100 nF, 50 V, X7R	8
D-TVS-MAIN	Main TVS diode	22 V–24 V standoff, high pulse power	1
R-BLEED	Bleeder resistor	10 k Ω , 0.25 W	1
LED-MAIN	Main power LED	suitable indicator LED	1
R-LED	LED resistor	6.8 k Ω –10 k Ω , 0.25 W	1

Table 10: PWM Signal Components

Ref.	Component	Rating / Value	Qty
J-PWM	IDC connector	2 by 8, 16-pin	1
R-PWM-1–8	Series resistor	100 Ω	8
R-PD-1–8	Pull-down resistor	10 k Ω	8
D-ESD	Optional ESD diode array	Low-capacitance signal ESD	1–2

- Cable-side main fuse is included near the battery.
- E-stop / contactor is upstream of PCB if used.
- Dual XT90 inputs are short and symmetric.
- Central node is a large filled polygon, not a trace.
- No long 200 A spine exists.
- All branch fuse inputs connect directly to the central node.
- Eight branch fuses are placed close to the node.
- Each branch fuse is rated 30 A.
- Each XT60 is behind its own branch fuse.
- Branch pours are 12 mm–15 mm where possible.
- Absolute branch bottlenecks are not below 8 mm–10 mm unless extremely short.
- High-current pads use direct copper connection.
- No thermal reliefs on XT90, fuse, XT60, TVS, or bulk capacitor pads.
- L1/L3 +VBAT copper is heavily stitched.
- L2/L4 GND copper is heavily stitched.
- GND is common and low impedance.
- TVS does not block or neck the main current path.
- PWM traces stay away from input, fuses, TVS, and central node.
- IDC pinout alternates PWM/GND.
- Cable strain relief is included for XT90 and XT60 wiring.
- Thermal camera test is planned.

15 TEST PLAN

15.1 Continuity Test

Before applying power:

- verify +VBAT to GND is not shorted,
- verify each branch positive passes through its branch fuse,
- verify each ESC negative returns to common GND,

- verify dual XT90 positive inputs join at the central node,
- verify dual XT90 negative inputs join at common GND,
- verify PWM lines are not shorted to +VBAT,
- verify IDC pinout.

15.2 Low-Current Power-Up

Use a current-limited bench supply:

$$V_{\text{supply}} = 16\text{ V} - 16.8\text{ V} \quad (50)$$

$$I_{\text{limit}} = 0.5\text{ A} - 1\text{ A} \quad (51)$$

Check:

- power LED turns on,
- no unexpected heating,
- capacitors charge normally,
- TVS does not heat during normal supply voltage.

15.3 Progressive Load Test

Test progressively:

1. 1 ESC at low throttle.
2. 1 ESC at high throttle.
3. 2 ESCs.
4. 4 ESCs.
5. 8 ESCs with controlled ramp.

Measure:

- input current,
- current through each XT90 input if possible,
- voltage drop from XT90 input to each XT60 connector,
- temperature of central node,
- temperature of branch fuse holders,
- temperature of XT60 pads,
- temperature of XT90 pads,
- TVS temperature,
- PWM stability during throttle changes.

Suggested pass criteria:

- no copper node above approximately 60°C – 70°C during expected operation,
- no connector or fuse holder hotspot significantly above surrounding copper,
- voltage drop from input to XT60 branch preferably below 100 mV – 200 mV under heavy load,
- dual XT90 inputs share current reasonably well,
- no PWM glitches during high-current throttle changes.

16 FINAL DESIGN SUMMARY

Design Note: For a compact 90 mm by 90 mm PCB, the single central star-node architecture is more practical than forcing two artificial banks. The safety condition is that the main fuse must be on the cable near the battery, while each ESC branch remains individually fused on the PCB.

Table 11: Recommended Parameters for Compact Central Star-Node Architecture

Item	Recommendation
Battery input	4S lithium, 14.8 V nominal, 16.8 V full
Architecture	One compact central 200 A star node with eight fused branches
Board size target	Approximately 90 mm by 90 mm
PCB	4-layer, 4 oz copper preferred
Stack-up	L1 +VBAT, L2 GND, L3 +VBAT, L4 GND + PWM/signal
Main input	Dual XT90-S or dual XT90, symmetric wiring required
Main fuse	Off-PCB cable-side fuse, 225 A–250 A
E-stop	Off-PCB contactor or manual isolator before PCB
Branch fuses	8× 30 A mini blade / ATO / ATC fuses
ESC outputs	XT60 preferred; XT30 only if derated and thermally validated
Central node	Short, wide copper polygon directly after dual XT90 input
Node length	Preferably below 20 mm–30 mm before branch split
Branch width	12 mm–15 mm per ESC branch; 8 mm–10 mm compact minimum
Vias	0.5 mm–0.6 mm drill / 1.0 mm–1.2 mm diameter preferred
Central node vias	100+ +VBAT vias distributed across central node
Branch vias	12–20 vias per branch per net
Main capacitance	2200 μ F–4700 μ F, 35 V or 50 V
Branch capacitance	8× 470 μ F–1000 μ F; 8× 1 μ F; 8× 100 nF
TVS	Main TVS, 22 V–24 V standoff, SMCJ or stronger
PWM connector	16-pin IDC, alternating PWM/GND
PWM passives	100 Ω series + 10 k Ω pulldown per PWM channel
ESC cooling	External aluminium heat block